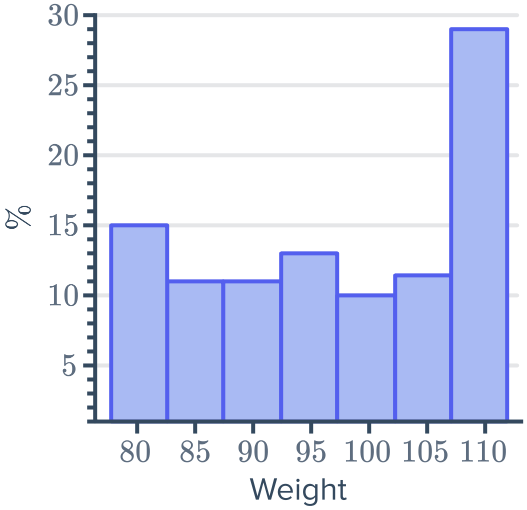
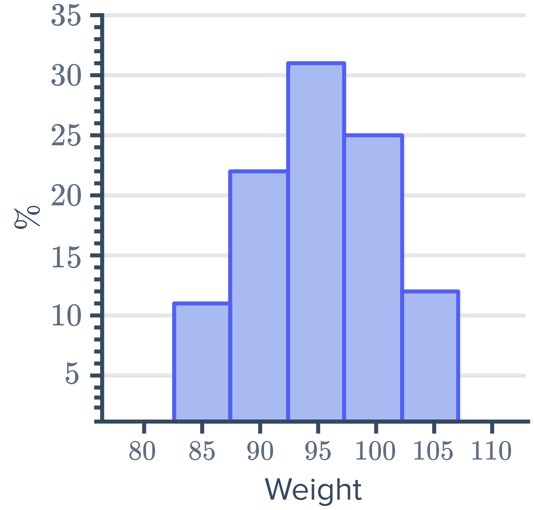
The normal distribution

1 State whether the following graphs of data are normally distributed:

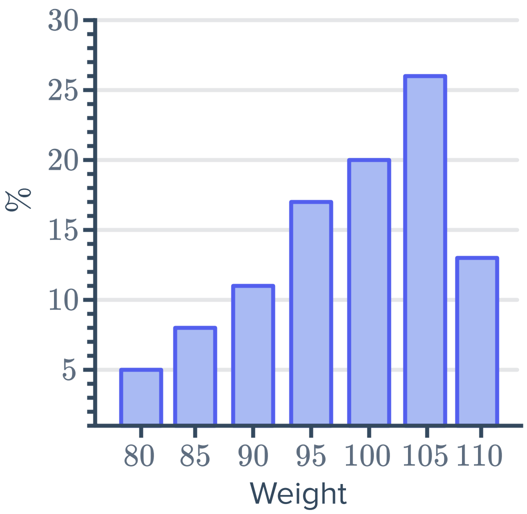
a



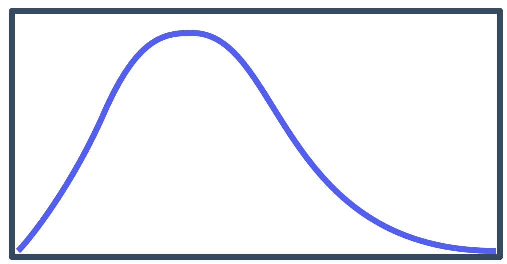
b



c



d



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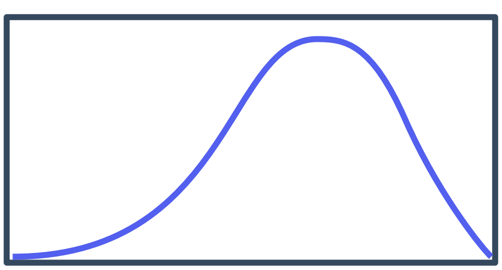

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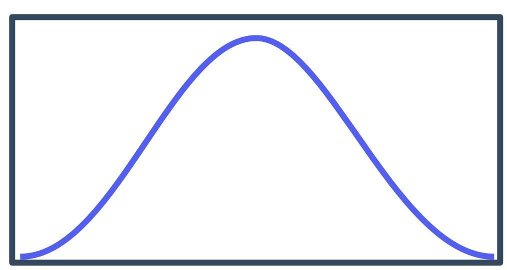

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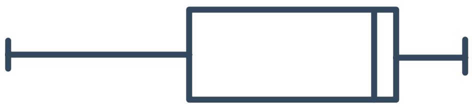
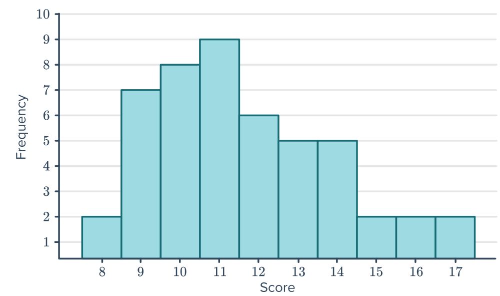
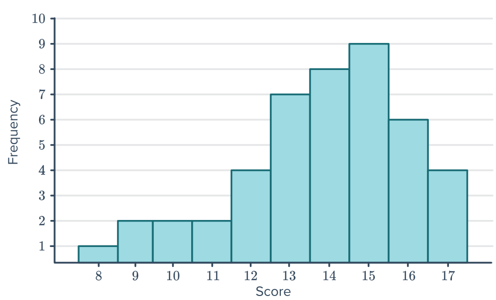


i

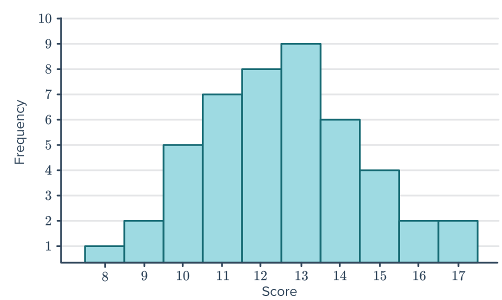
j



k



l

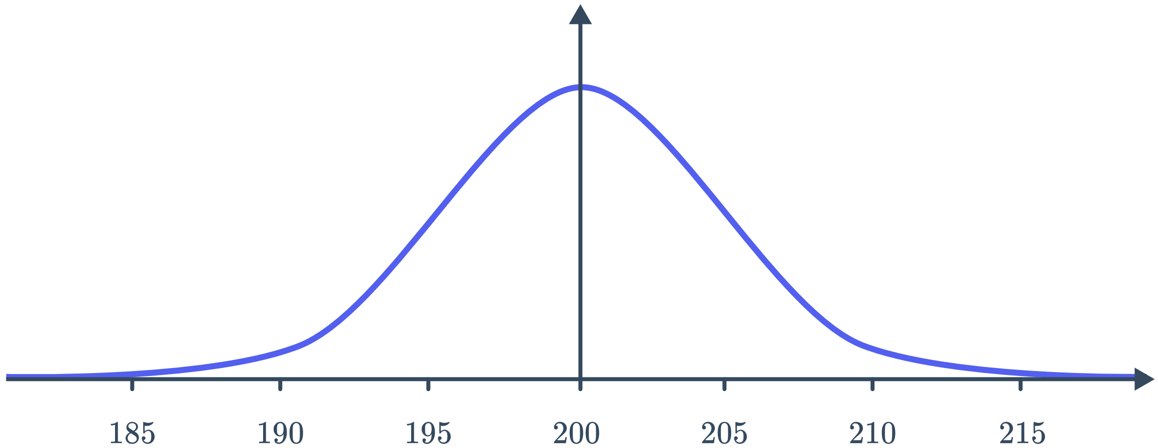


2 Consider the given normal distribution:

a State the mean.

b State the median.

c State the mode.



3 State whether the following statements are true of the normal distribution curve:

a There are the same number of scores above and below the mean.

b The curve is symmetrical.

c The fewest scores lie around the mean.

d The curve is asymmetrical.

e The spread of the normal distribution changes depending on the mean.

f A greater mean will result in a skewed curve.

g The mean, median and mode all have the same value.

h The spread of the normal distribution changes depending on the standard deviation.

4 Given a normal distribution, state the percentage of people that lie:

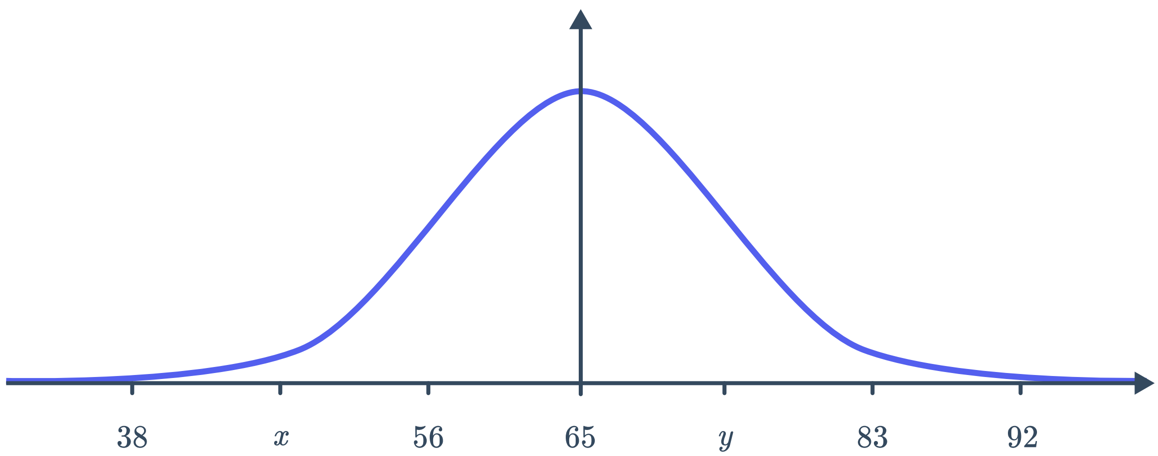
a Above the mean.

b Below the mean.

5 A data set has a mean of 80 and a standard deviation of 3. Find the score that is 3 standard deviations above the mean.

Applications

6 The results from an exam were approximately normally distributed. The mean score was 65, with a standard deviation of 9. The points marked on the horizontal axis are separated by 1 standard deviation:



a State the value of x on the graph.

b State the value of y on the graph.

7 In a male population, the mean height was 175 cm, with a standard deviation of 7 cm.

a State the height of Bob who is 4 standard deviations above the mean.

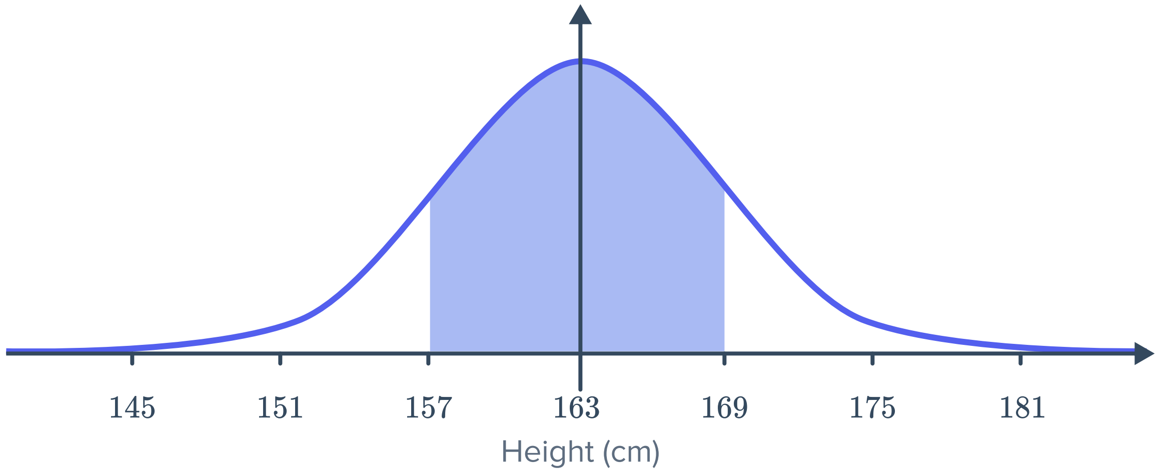
b State the height of John who is 2 standard deviations below the mean.

8 A sample of professional basketball players is normally distributed and gives the mean height as 199 cm with a standard deviation of 10 cm. State the height of a basketball player who is:

a 3 standard deviations above the mean

b 1.5 standard deviations below the mean

9 The diagram below shows the distribution o[redacted]ales heights in a population:



a If 34% of females lie between $157 \mathrm{~cm}$ and $163 \mathrm{~cm}$, what percentage of females lie between $163 \mathrm{~cm}$ and $169 \mathrm{~cm}$.

b What length is one standard deviation equal to?

10 Assume the mass of sumo wrestlers is normally distributed, with a mean mass of $158 \mathrm{~kg}$ and a standard deviation of $10 \mathrm{~kg}$.

a What percentage of wrestler would weigh more than $158 \mathrm{~kg}$?

b A sumo wrestler's weight is one standard deviation below the mean. How much does he weigh?